SigLIP · arXiv 2303.15343 · Sep 14

Empiricist

seat 4 · 11+3

Small-BS sigmoid vs softmax · bias · probe

Paper club · VLA Architectures · dark academic

H · Empiricist hypothesis (P0)

At BS ≤ 1k, sigmoid + learned bias trains faster / stabler retrieval than matched softmax CLIP; removing bias hurts early stability.

Success criterion

Qualitative Fig. 2 gap — NOT paper ImageNet %

Experiment plan

Pri	ID	Experiment	Notes
P0	E1	Sigmoid vs softmax	Tiny paired set · BS ≤ 1k
P0	E2	Bias b=−10 ablation	w/ vs w/o · early stability
P0	E3	Frozen probe	SigLIP vs CLIP on household/robot stills
P1	E4	Encoder-swap outline	OpenPI/$\pi_{0.5}$ · design only
Skip	—	Full WebLI reproduce	Not honest on user cluster

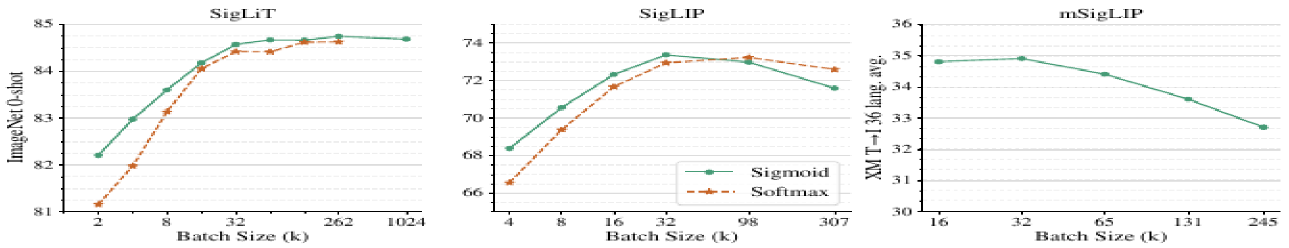

Figure 2: The effect of pre-training batch size. **Left: SigLiT results**, trained for 18B seen examples. Sigmoid loss outperforms the softmax loss significantly with small batch sizes, and performs similarly at larger batch sizes. We successfully trained an SigLiT model with up to *one million* batch size. However, performance for both sigmoid and softmax saturate at around 32 k batch size. **Middle: SigLIP results**, trained for 9B seen examples. Both sigmoid loss and softmax loss saturate at a reasonable batch size, while the peak of the sigmoid loss comes earlier and slightly outperforms the peak of the softmax loss. A very large batch size hurts both losses. **Right: mSigLIP results**, trained for 30B seen examples. With a multilingual setup using over 100 languages, 32 k batch size is surprisingly sufficient and scaling beyond that hurts performance on a 36-language cross-modal retrieval task.

method. In words, we first compute the component of the loss corresponding to the positive pairs, and $b-1$ negative pairs. We then permute representations across devices, so each device takes negatives from its neighbouring device (next iteration of sum **B**). The loss is then calculated with respect to this chunk (sum **C**). This is done independently in each device, such that each device computes the loss with respect to its local batch b. Losses can then simply be summed across all devices (sum **A**). Individual collective permutes (for sum **B**) are fast (and indeed D collective permutes is typically faster than two all-gathers between D devices), and the memory cost at any given moment is reduced from $|\mathcal{B}|^2$ to b^2 (for sum **C**). Usually b is constant as scaling $|\mathcal{B}|$ is achieved by increasing the number of accelerators. Due to being quadratic with respect to the batch size, the vanilla loss computation rapidly bottlenecks scaling up. This chunked approach enabled training with batch sizes over 1 million on relatively few devices.

4. Results

In this section, we evaluate the proposed SigLiT and SigLIP models across a wide range of batch sizes. We discuss what can be achieved with a small number of accelerator chips, using both SigLiT and SigLIP recipes. We also briefly discuss the impact of batch size on multilingual language image pre-training. We ablate the importance of our large-batch stabilization modification and the introduced learned bias term and present a study on the effect of positive and negative pairs ratio in the sigmoid loss. Lastly, we explore SigLIP's data noise robustness.

To validate our models, we report zero-shot transfer results on the ImageNet dataset [14] and zero-shot retrieval results across 36 languages on the XM3600 dataset [44]. We use the ScalingViT-Adafactor optimizer [58] by default for all our experiments.

4.1. SigLiT: Scaling batch size to the limit

Following [59], we use the same precomputed embeddings for the images using a ViT-g vision model, and train a base size text tower from scratch with the same hyperparameters using the LiT image-text dataset [59].

We perform a study over a wide range of batch sizes, from 512 to 1 M, demonstrating the impact of batch size for contrastive learning. Results are presented in Figure 2 (left). When the batch size is smaller than 16 k, sigmoid loss outperforms softmax loss by a large margin. With growing batch sizes, we observe that softmax loss quickly catches up and potentially slightly underperforms sigmoid loss with a large enough batch size. Overall, we recommend using the SigLIP recipe for large batch sizes as well, due to the simplicity, compute savings, and straightforward memory efficient implementation.

There is a consensus that contrastive learning benefits from large batch sizes, while most of the existing studies stop at 64 k batch size [59, 35, 10]. We successfully trained an SigLiT model at one million batch size, to explore the limit of contrastive learning. To our surprise, the performance saturates at 32 k batch size, further scaling up the batch size only gives a minor boost, and the model peaks at

Target qualitative shape — Fig. 2 (not absolute INet %)

GPU overnight budget

E1 / E2

1×24–40GB
6–24 h
ViT-S + tiny text
BS 64–1k

E3 probe

1×16–24GB
1–4 h
frozen features
linear / NN

E4 outline

multi-A100 days
NOT this cycle
design spike only

Honesty line

Paper’s 84.5% / 73.4% need TPU-v4 fleet + WebLI.
Local success = qualitative small-BS gap + probe implications.

Success vs paper claims

Paper: sigmoid > softmax @ small BS	Local: clear R@1 gap or faster early rise @ BS≤1k
Paper: bias b stabilizes	Local: w/o b → early spike / slower R@1
Paper: ~32k sweet spot	Discuss only — cite Fig. 2; no local BS=32k
Paper: 84.5% / 73.4% INet	Cite as literature upper bound — not expected
Downstream encoder quality	E3: SigLIP ≥ CLIP on household/robot stills

Loss sketch · pairwise sigmoid

$$L = -1/|B| \ \Sigma_{\{i,j\}} \ \log \sigma(z_{ij} \cdot (t \cdot x_i^{\top} y_j + b))$$

$z_{ij} = +1$ match · $z_{ij} = -1$ non-match · $t = \exp(t')$ · $b \approx -10$ init

No softmax denom

Each pair is an independent logistic term — no batch NCE

Bias is load-bearing

Extreme neg:pos ratio needs $b \approx -10$ so early grads stay sane

Chunked systems

Permute text shards; only b_{local}^2 logits in memory (Fig.1)

Sigmoid vs Softmax · the surgical swap

Spine of this paper: pairwise logistic replaces batch InfoNCE

CLIP Softmax (InfoNCE)

- ▸ Asymmetric i→t + t→i softmax
- ▸ Needs full-batch normalization
- ▸ All-gather / sync across devices
- ▸ Negatives compete exclusively
- ▸ Weak at small batch sizes
- ▸ Memory ~ $|B|^2$ logits + gather

SigLIP Pairwise Sigmoid

- ▸ Independent +1 / −1 per pair
- ▸ No global denom / no all-gather
- ▸ Chunked distributed loss (Fig.1)
- ▸ Learnable t + bias b (b≈−10)
- ▸ Strong at BS ≤ 1k–16k
- ▸ Saturates ~32k (larger can hurt)

E4 outline only

OpenPI encoder-swap = design spike · not overnight Empiricist job

Empiricist close

H @ BS≤1k · E1/E2/E3 · success = Fig.2 gap · skip WebLI